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COMPUTER NETWORKS

TEAM NETWORKS

Department of Computer Science and Engineering

Unit – 4 Network Layer and Internet Protocol

4.5 Introduction to Network Layer Protocols

- DHCP
- ICMP

4.6 IPv6 Addressing

4.7 Introduction to Routing Algorithms

Distance vector algorithm

Based on *Bellman-Ford* (BF) equation (dynamic programming):

Bellman-Ford equation

Let $D_x(y)$: cost of least-cost path from x to y .

Then:

$$D_x(y) = \min_v \{ c_{x,v} + D_v(y) \}$$

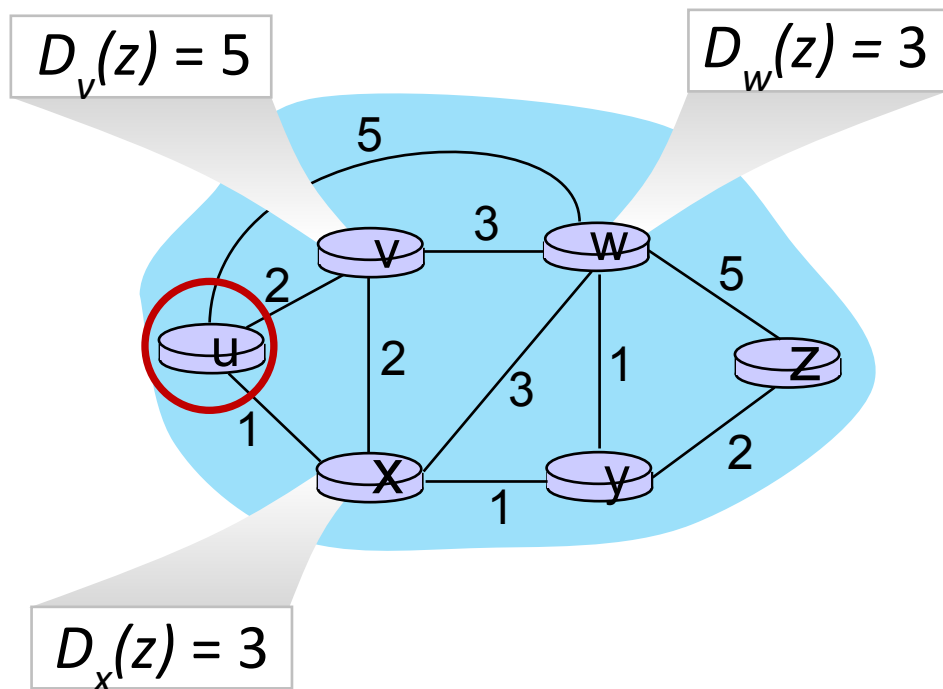
min taken over all neighbors v of x

direct cost of link from x to v

v 's estimated least-cost-path cost to y

Bellman-Ford: Example

Suppose that u 's neighboring nodes, x, v, w , know that for destination z :



Bellman-Ford equation says:

$$\begin{aligned}
 D_u(z) &= \min \{ c_{u,v} + D_v(z), \\
 &\quad c_{u,x} + D_x(z), \\
 &\quad c_{u,w} + D_w(z) \} \\
 &= \min \{ 2 + 5, \\
 &\quad 1 + 3, \\
 &\quad 5 + 3 \} = 4
 \end{aligned}$$

node achieving minimum (x) is next hop on estimated least-cost path to destination (z)

Distance vector algorithm:

key idea:

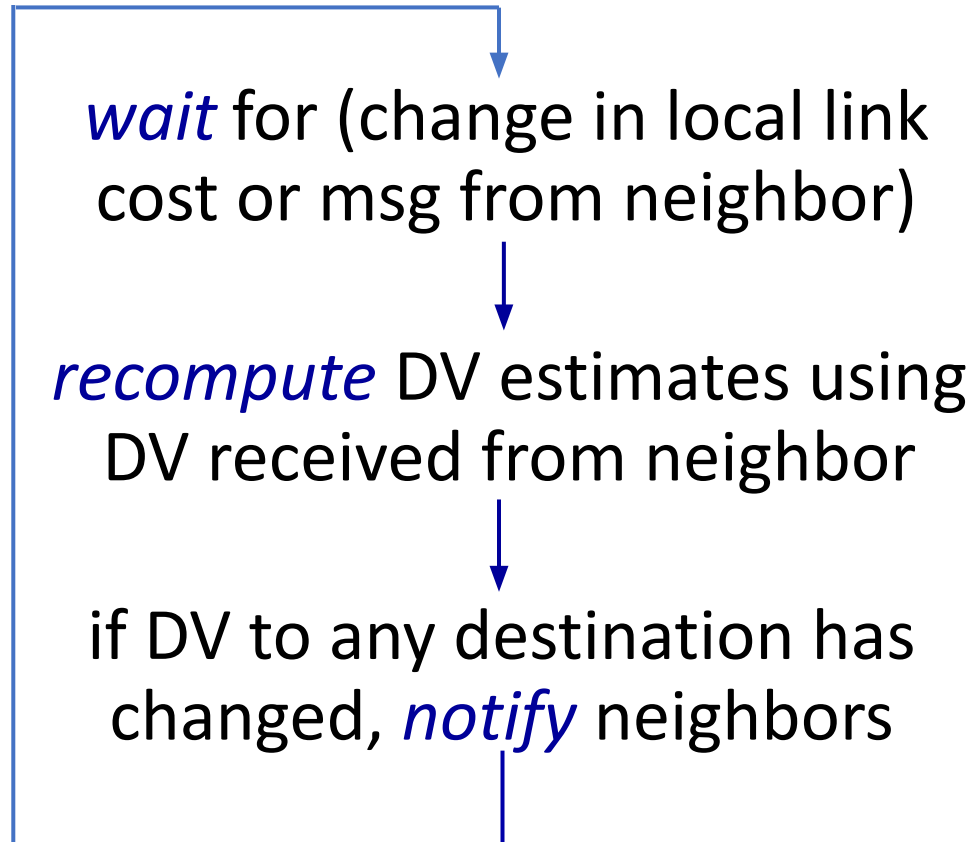
- from time-to-time, each node sends its own distance vector estimate to neighbors
- when x receives new DV estimate from any neighbor, it updates its own DV using B-F equation:

$$D_x(y) \leftarrow \min_v \{c_{x,v} + D_v(y)\} \text{ for each node } y \in N$$

- under minor, natural conditions, the estimate $D_x(y)$ converge to the actual least cost $d_x(y)$

Distance vector algorithm:

each node:



iterative, asynchronous: each local iteration caused by:

- local link cost change
- DV update message from neighbor

distributed, self-stopping: each node notifies neighbors *only* when its DV changes

- neighbors then notify their neighbors – *only if necessary*
- no notification received, no actions taken!

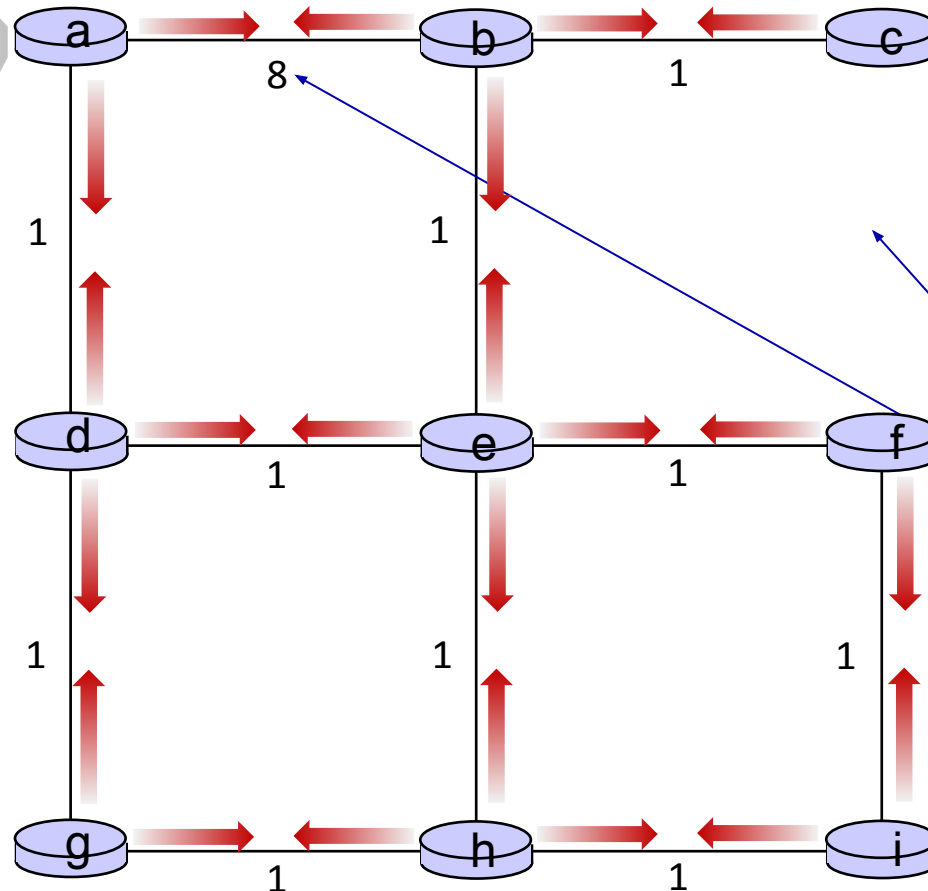
Distance vector: example



$t=0$

- All nodes have distance estimates to nearest neighbors (only)
- All nodes send their local distance vector to their neighbors

DV in a:	
$D_a(a)=0$	
$D_a(b)=8$	
$D_a(c)=\infty$	
$D_a(d)=1$	
$D_a(e)=\infty$	
$D_a(f)=\infty$	
$D_a(g)=\infty$	
$D_a(h)=\infty$	
$D_a(i)=\infty$	



A few asymmetries:

- missing link
- larger cost

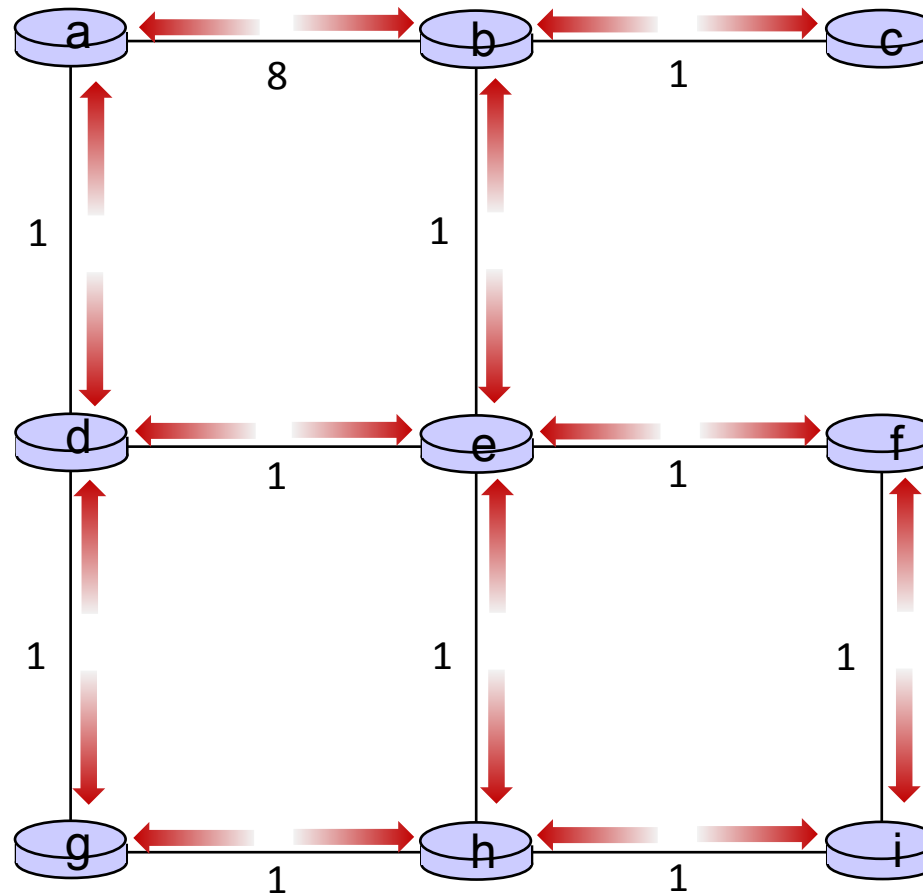
Distance vector: Iteration



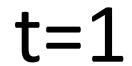
$t=1$

All nodes:

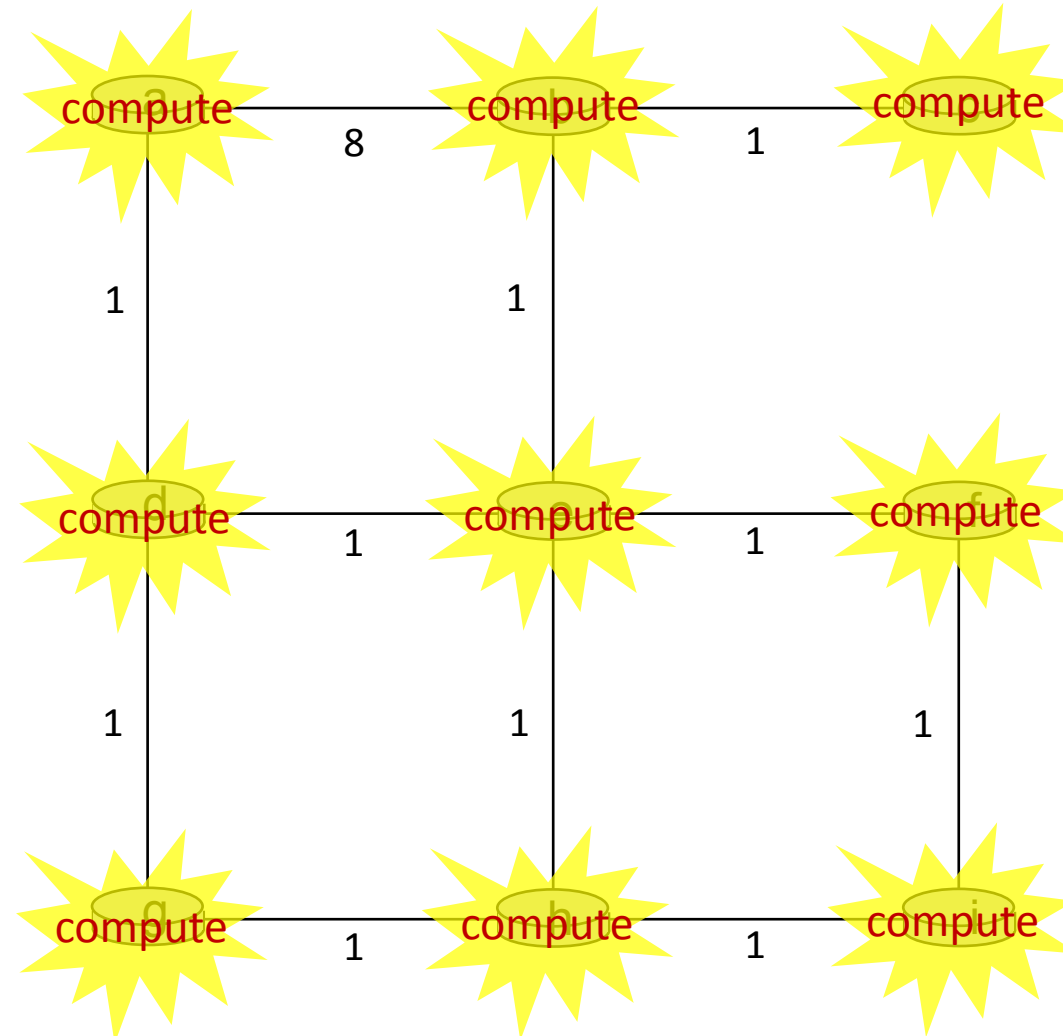
- receive distance vectors from neighbors
- compute their new local distance vector
- send their new local distance vector to neighbors



Distance vector: Iteration



- receive distance vectors from neighbors
- **compute their new local distance vector**
- send their new local distance vector to neighbors



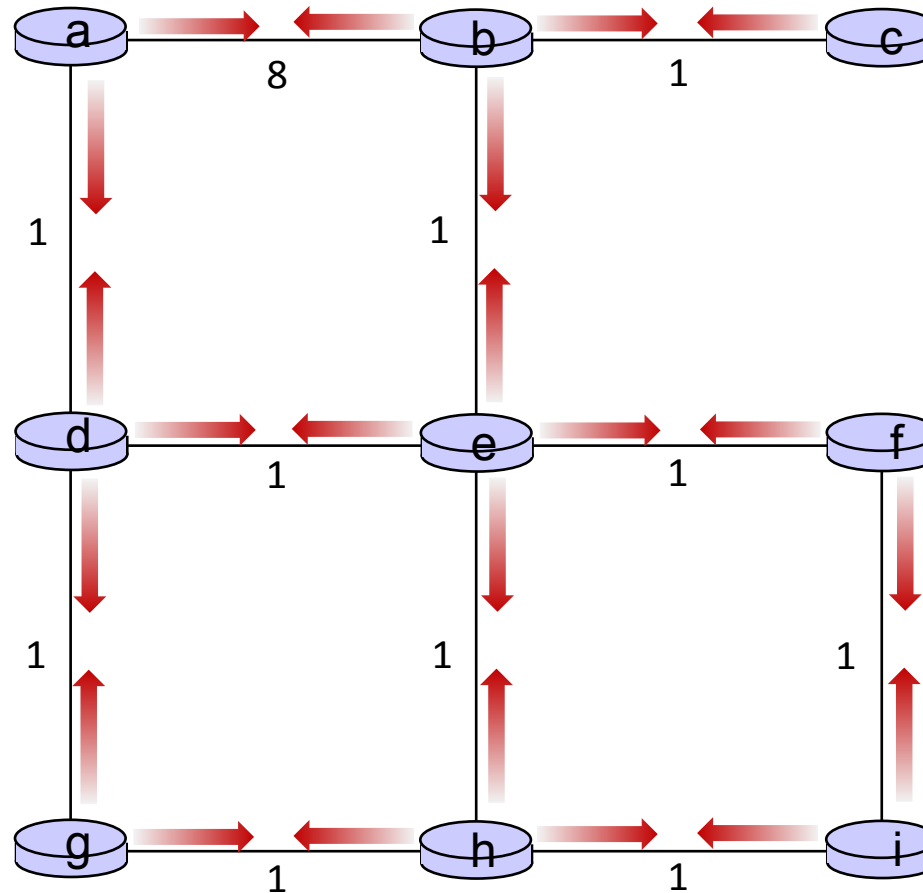
Distance vector: Iteration



$t=1$

All nodes:

- receive distance vectors from neighbors
- compute their new local distance vector
- send their new local distance vector to neighbors



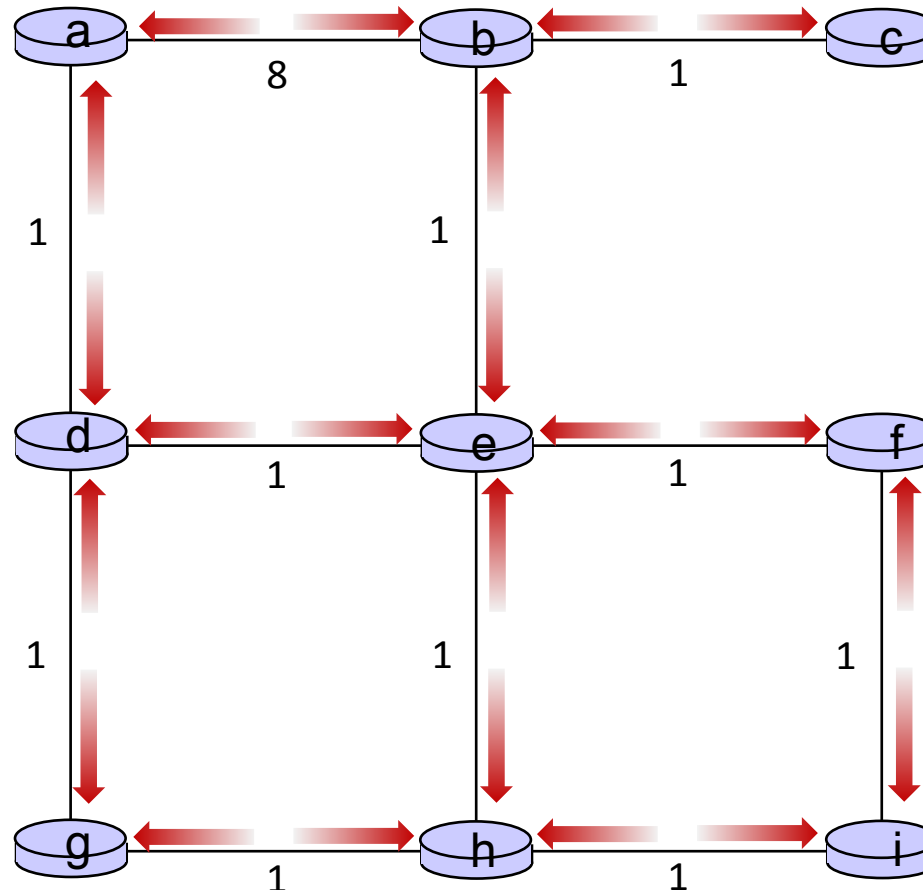
Distance vector: Iteration



$t=2$

All nodes:

- receive distance vectors from neighbors
- compute their new local distance vector
- send their new local distance vector to neighbors



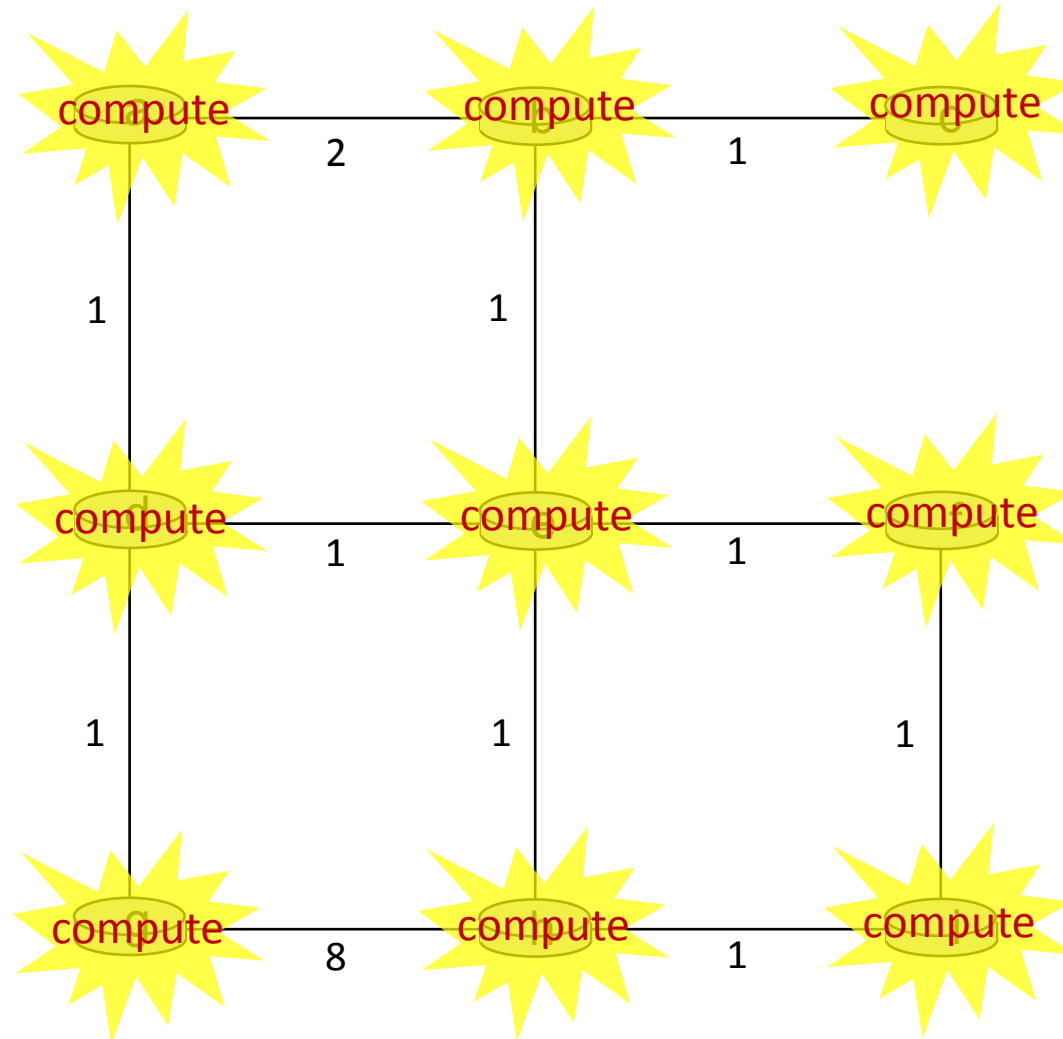
Distance vector: Iteration



t=2

All nodes:

- receive distance vectors from neighbors
- compute their new local distance vector
- send their new local distance vector to neighbors



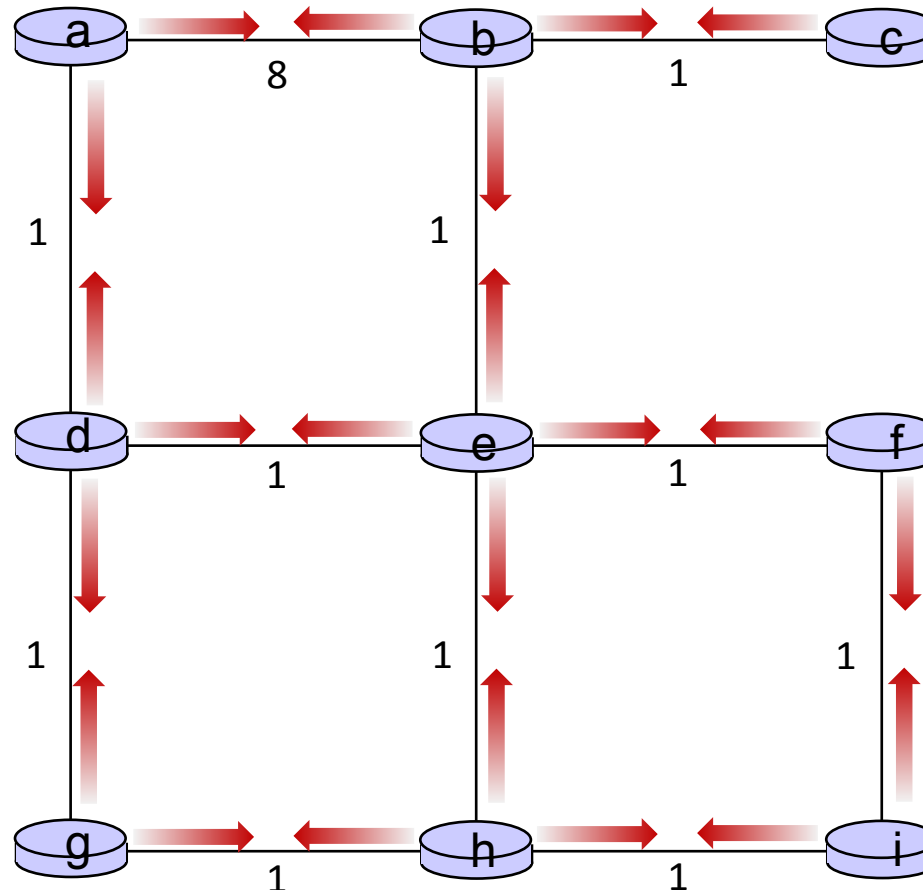
Distance vector: Iteration



$t=2$

All nodes:

- receive distance vectors from neighbors
- compute their new local distance vector
- send their new local distance vector to neighbors



.... and so on

Distance vector example: computation



t=1

- b receives DVs from a, c, e

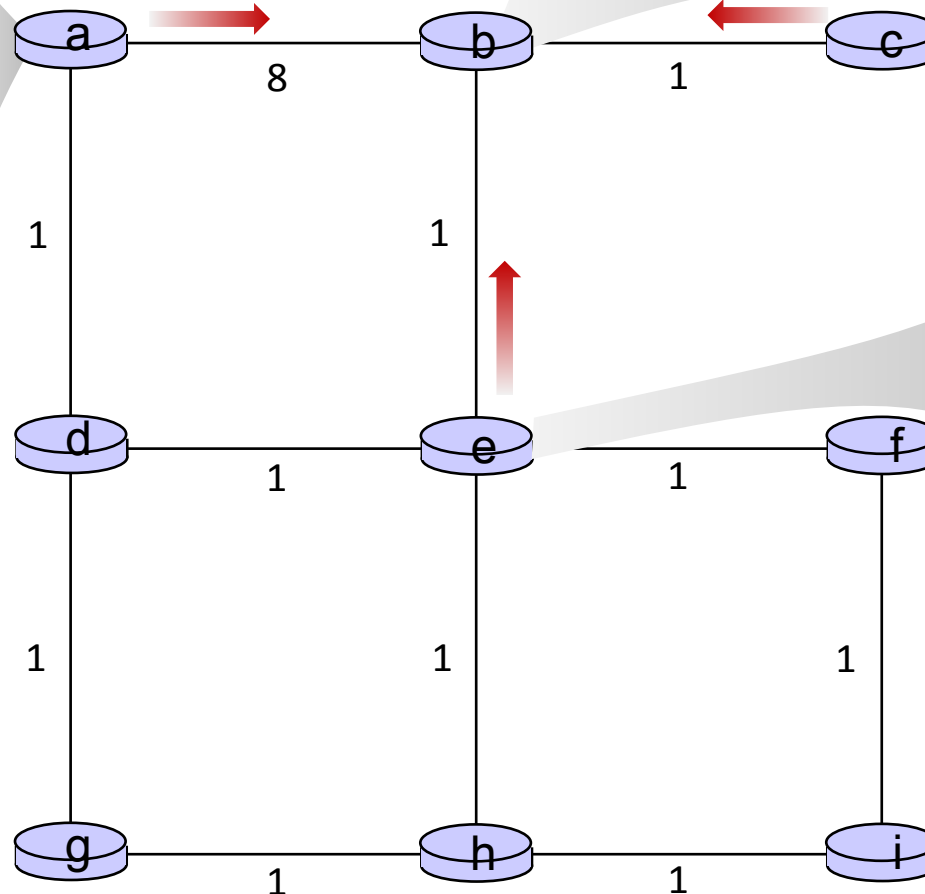
Let's next take a look at the iterative *computations* at nodes

DV in a:
$D_a(a) = 0$
$D_a(b) = 8$
$D_a(c) = \infty$
$D_a(d) = 1$
$D_a(e) = \infty$
$D_a(f) = \infty$
$D_a(g) = \infty$
$D_a(h) = \infty$
$D_a(i) = \infty$

DV in b:	
$D_b(a) = 8$	$D_b(f) = \infty$
$D_b(c) = 1$	$D_b(g) = \infty$
$D_b(d) = \infty$	$D_b(h) = \infty$
$D_b(e) = 1$	$D_b(i) = \infty$

DV in c:
$D_c(a) = \infty$
$D_c(b) = 1$
$D_c(c) = 0$
$D_c(d) = \infty$
$D_c(e) = \infty$
$D_c(f) = \infty$
$D_c(g) = \infty$
$D_c(h) = \infty$
$D_c(i) = \infty$

DV in e:
$D_e(a) = \infty$
$D_e(b) = 1$
$D_e(c) = \infty$
$D_e(d) = 1$
$D_e(e) = 0$
$D_e(f) = 1$
$D_e(g) = \infty$
$D_e(h) = 1$
$D_e(i) = \infty$





$t=1$

- b receives DVs from a, c, e, computes:

$$D_b(a) = \min\{c_{b,a} + D_a(a), c_{b,c} + D_c(a), c_{b,e} + D_e(a)\} = \min\{8, \infty, \infty\} = 8$$

$$D_b(c) = \min\{c_{b,a} + D_a(c), c_{b,c} + D_c(c), c_{b,e} + D_e(c)\} = \min\{\infty, 1, \infty\} = 1$$

$$D_b(d) = \min\{c_{b,a} + D_a(d), c_{b,c} + D_c(d), c_{b,e} + D_e(d)\} = \min\{9, 2, \infty\} = 2$$

$$D_b(e) = \min\{c_{b,a} + D_a(e), c_{b,c} + D_c(e), c_{b,e} + D_e(e)\} = \min\{\infty, \infty, 1\} = 1$$

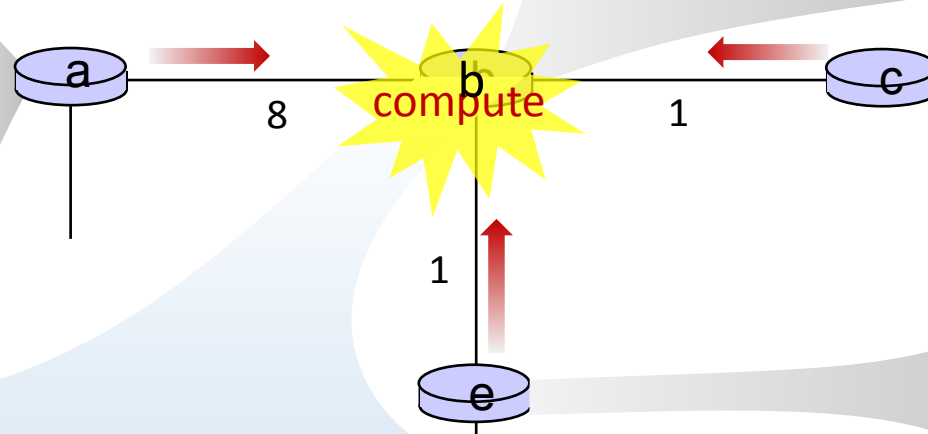
$$D_b(f) = \min\{c_{b,a} + D_a(f), c_{b,c} + D_c(f), c_{b,e} + D_e(f)\} = \min\{\infty, \infty, 2\} = 2$$

$$D_b(g) = \min\{c_{b,a} + D_a(g), c_{b,c} + D_c(g), c_{b,e} + D_e(g)\} = \min\{\infty, \infty, \infty\} = \infty$$

$$D_b(h) = \min\{c_{b,a} + D_a(h), c_{b,c} + D_c(h), c_{b,e} + D_e(h)\} = \min\{\infty, \infty, 2\} = 2$$

$$D_b(i) = \min\{c_{b,a} + D_a(i), c_{b,c} + D_c(i), c_{b,e} + D_e(i)\} = \min\{\infty, \infty, \infty\} = \infty$$

DV in a:
$D_a(a) = 0$
$D_a(b) = 8$
$D_a(c) = \infty$
$D_a(d) = 1$
$D_a(e) = \infty$
$D_a(f) = \infty$
$D_a(g) = \infty$
$D_a(h) = \infty$
$D_a(i) = \infty$



DV in b:

$D_b(a) = 8$	$D_b(f) = \infty$
$D_b(c) = 1$	$D_b(g) = \infty$
$D_b(d) = \infty$	$D_b(h) = \infty$
$D_b(e) = 1$	$D_b(i) = \infty$

DV in c:
$D_c(a) = \infty$
$D_c(b) = 1$
$D_c(c) = 0$
$D_c(d) = \infty$
$D_c(e) = \infty$
$D_c(f) = \infty$
$D_c(g) = \infty$
$D_c(h) = \infty$
$D_c(i) = \infty$

DV in e:
$D_e(a) = \infty$
$D_e(b) = 1$
$D_e(c) = \infty$
$D_e(d) = 1$
$D_e(e) = 0$
$D_e(f) = 1$
$D_e(g) = \infty$
$D_e(h) = 1$
$D_e(i) = \infty$

DV in b:

$$D_b(a) = 8$$

$$D_b(f) = 2$$

$$D_b(c) = 1$$

$$D_b(g) = \infty$$

$$D_b(d) = 2$$

$$D_b(h) = 2$$

$$D_b(e) = 1$$

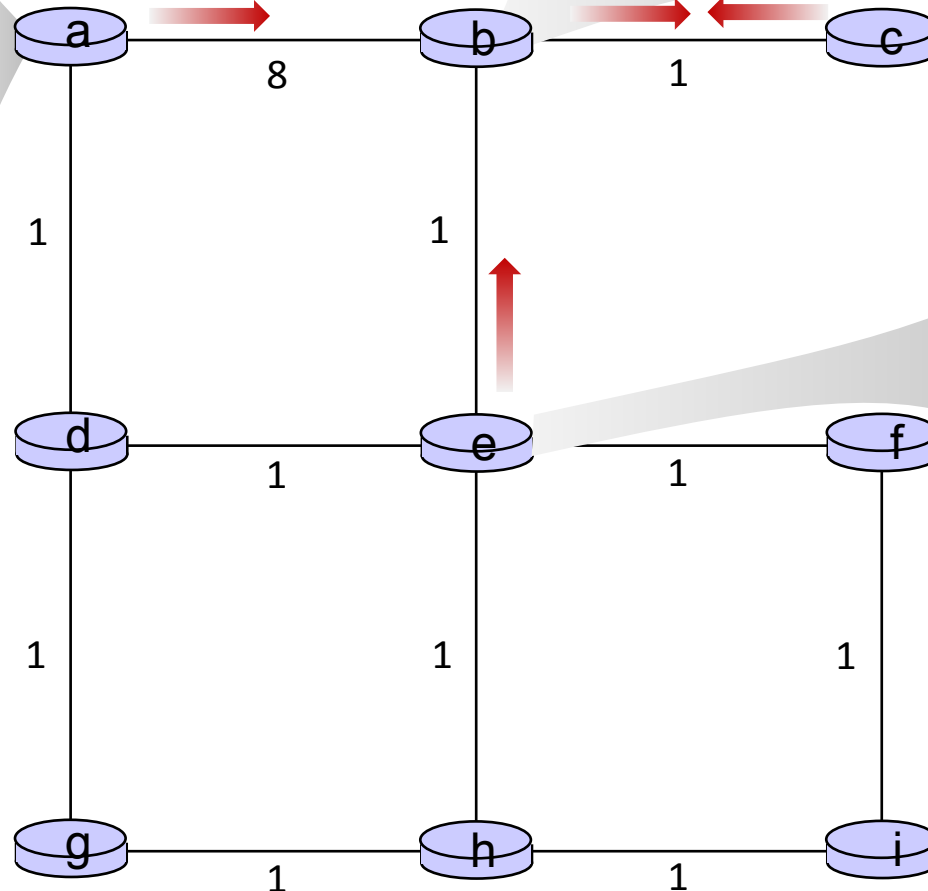
$$D_b(i) = \infty$$



$t=1$

- c receives DVs from b

DV in a:
$D_a(a)=0$
$D_a(b)=8$
$D_a(c)=\infty$
$D_a(d)=1$
$D_a(e)=\infty$
$D_a(f)=\infty$
$D_a(g)=\infty$
$D_a(h)=\infty$
$D_a(i)=\infty$



DV in b:	
$D_b(a)=8$	$D_b(f)=\infty$
$D_b(c)=1$	$D_b(g)=\infty$
$D_b(d)=\infty$	$D_b(h)=\infty$
$D_b(e)=1$	$D_b(i)=\infty$

DV in c:
$D_c(a)=\infty$
$D_c(b)=1$
$D_c(c)=0$
$D_c(d)=\infty$
$D_c(e)=\infty$
$D_c(f)=\infty$
$D_c(g)=\infty$
$D_c(h)=\infty$
$D_c(i)=\infty$

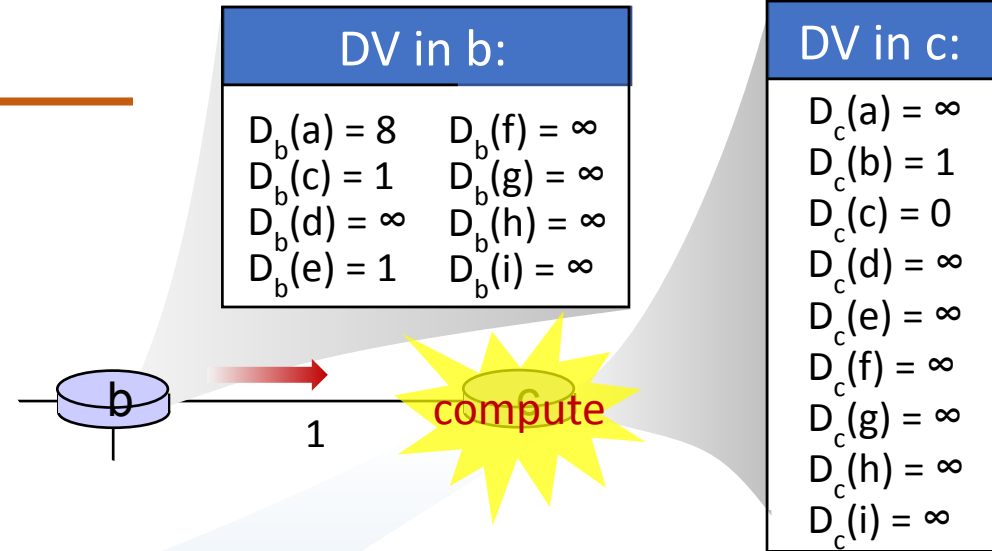
DV in e:
$D_e(a)=\infty$
$D_e(b)=1$
$D_e(c)=\infty$
$D_e(d)=1$
$D_e(e)=0$
$D_e(f)=1$
$D_e(g)=\infty$
$D_e(h)=1$
$D_e(i)=\infty$



t=1

- c receives DVs from b computes:

$$\begin{aligned}D_c(a) &= \min\{c_{c,b} + D_b(a)\} = 1 + 8 = 9 \\D_c(b) &= \min\{c_{c,b} + D_b(b)\} = 1 + 0 = 1 \\D_c(d) &= \min\{c_{c,b} + D_b(d)\} = 1 + \infty = \infty \\D_c(e) &= \min\{c_{c,b} + D_b(e)\} = 1 + 1 = 2 \\D_c(f) &= \min\{c_{c,b} + D_b(f)\} = 1 + \infty = \infty \\D_c(g) &= \min\{c_{c,b} + D_b(g)\} = 1 + \infty = \infty \\D_c(h) &= \min\{c_{c,b} + D_b(h)\} = 1 + \infty = \infty \\D_c(i) &= \min\{c_{c,b} + D_b(i)\} = 1 + \infty = \infty\end{aligned}$$



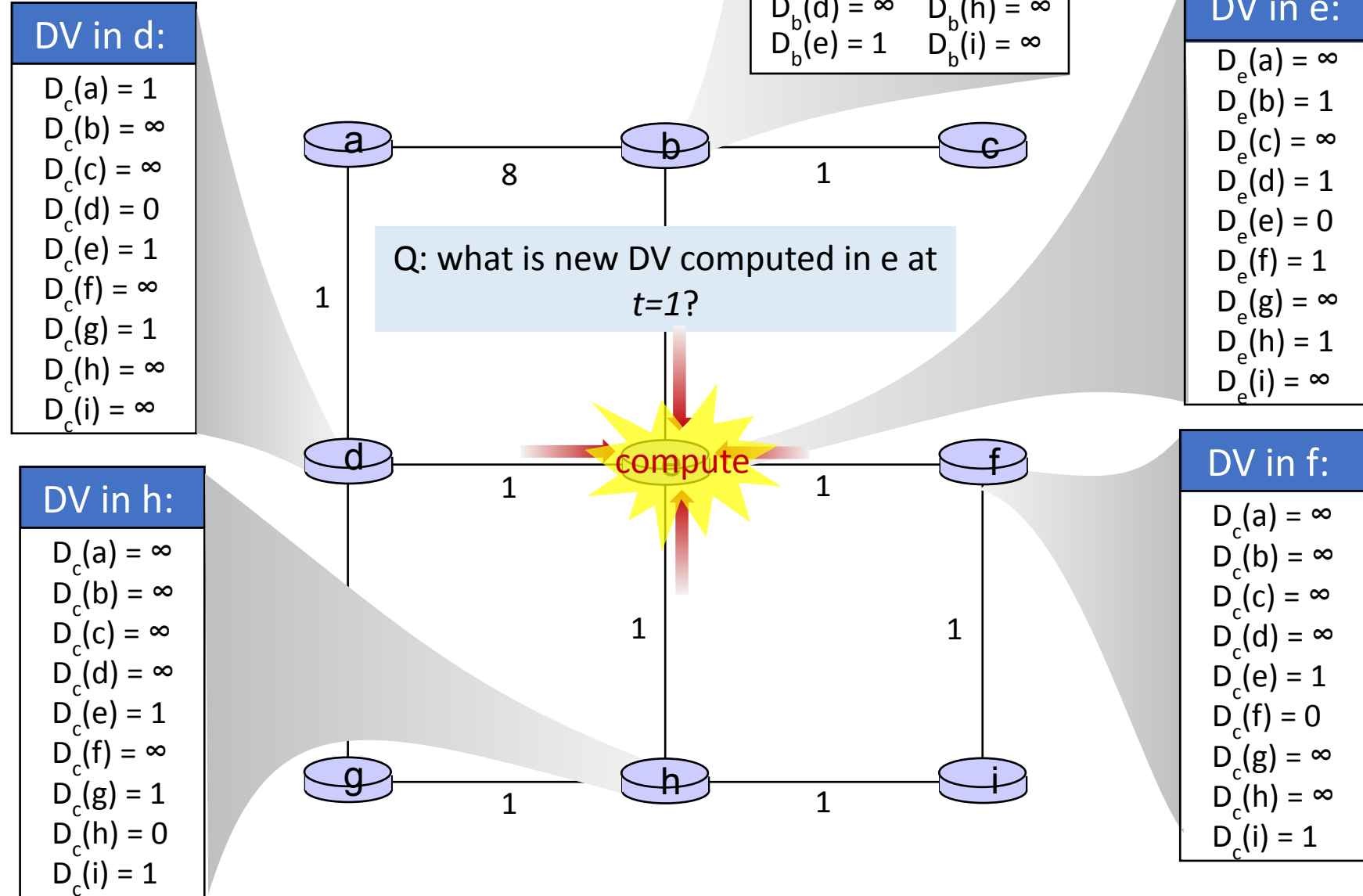
DV in c:	
$D_c(a) = 9$	
$D_c(b) = 1$	
$D_c(c) = 0$	
$D_c(d) = 2$	
$D_c(e) = \infty$	
$D_c(f) = \infty$	
$D_c(g) = \infty$	
$D_c(h) = \infty$	
$D_c(i) = \infty$	

* Check out the online interactive exercises for more examples:
http://gaia.cs.umass.edu/kurose_ross/interactive/



$t=1$

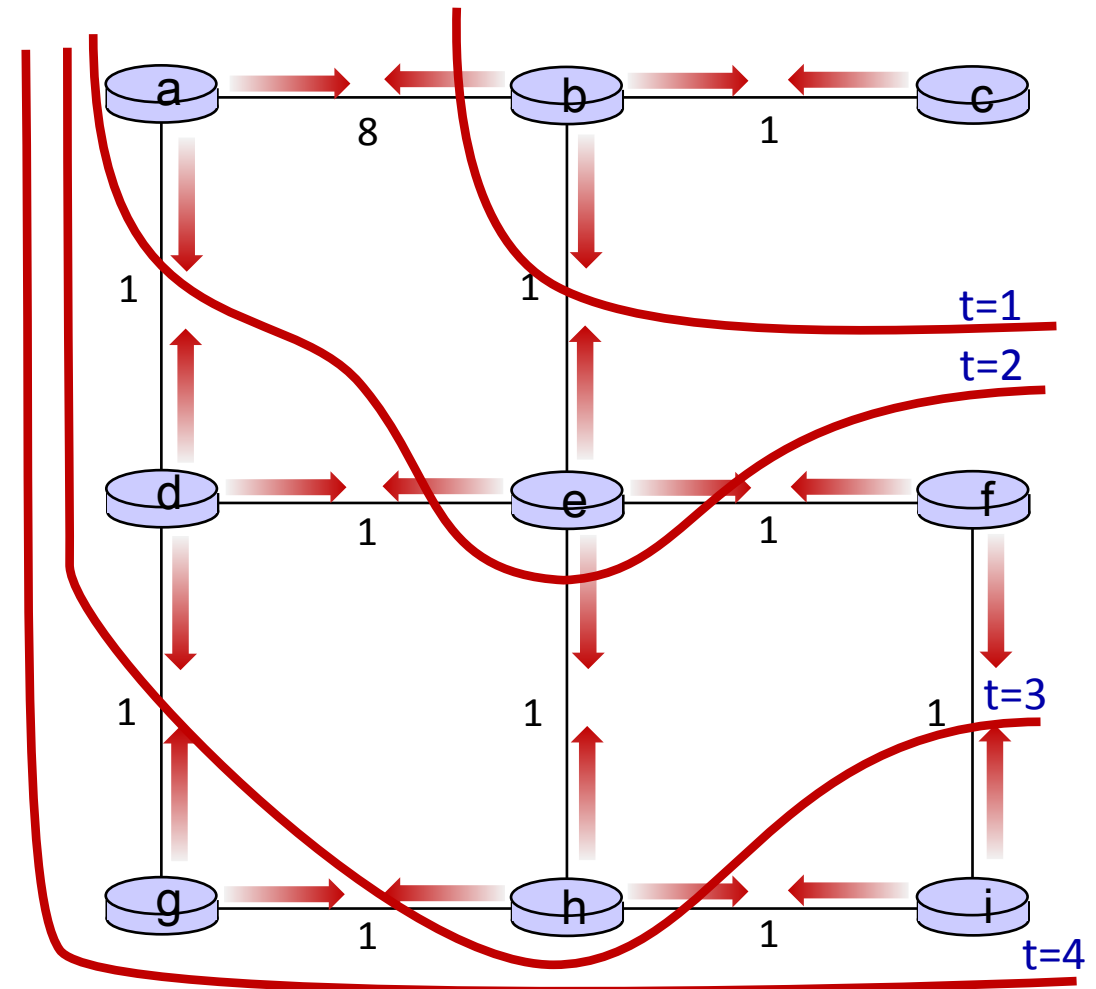
- e receives DVs from b, d, f, h



Distance vector: state information diffusion

Iterative communication, computation steps diffuses information through network:

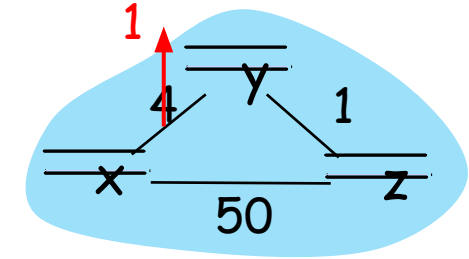
-  $t=0$ c's state at $t=0$ is at c only
-  $t=1$ c's state at $t=0$ has propagated to b, and may influence distance vector computations up to **1** hop away, i.e., at b
-  $t=2$ c's state at $t=0$ may now influence distance vector computations up to **2** hops away, i.e., at b and now at a, e as well
-  $t=3$ c's state at $t=0$ may influence distance vector computations up to **3** hops away, i.e., at b,a,e and now at c,f,h as well
-  $t=4$ c's state at $t=0$ may influence distance vector computations up to **4** hops away, i.e., at b,a,e, c, f, h and now at g,i as well



Distance vector: link cost changes

link cost changes:

- node detects local link cost change
- updates routing info, recalculates local DV
- if DV changes, notify neighbors



“good news
travels fast”

t_0 : y detects link-cost change, updates its DV, informs its neighbors.

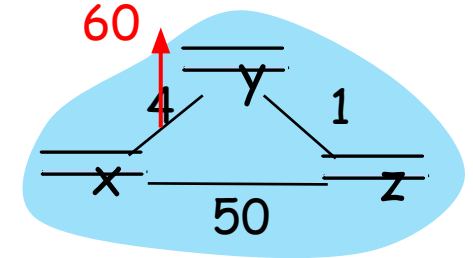
t_1 : z receives update from y , updates its table, computes new least cost to x , sends its neighbors its DV.

t_2 : y receives z 's update, updates its distance table. y 's least costs do *not* change, so y does *not* send a message to z .

Distance vector: link cost changes

link cost changes:

- node detects local link cost change
- “bad news travels slow” – count-to-infinity problem:
 - y sees direct link to x has new cost 60, but z has said it has a path at cost of 5. So y computes “my new cost to x will be 6, via z); notifies z of new cost of 6 to x.
 - z learns that path to x via y has new cost 6, so z computes “my new cost to x will be 7 via y), notifies y of new cost of 7 to x.
 - y learns that path to x via z has new cost 7, so y computes “my new cost to x will be 8 via y), notifies z of new cost of 8 to x.
 - z learns that path to x via y has new cost 8, so z computes “my new cost to x will be 9 via y), notifies y of new cost of 9 to x.
 - ...
- see text for solutions. *Distributed algorithms are tricky!*



	Distance Vector	Link State
Primary principle	Send entire routing table to its neighbors	Only provides link state information
Learning about network	Learn about network only from neighbors	Learn about network from all routers
Building the routing table	Based on inputs from only neighbors	Based on complete database collected from all routers
Advertisement of updates	Sends periodic updates every 30-90 seconds – Broadcasts updates	Use triggered updates, only when there is a change – Multicasts updates
Routing loops	Vulnerable	Less prone to routing loops

	Distance vector	Link State
Convergence (stabilization)	Slow	Fast
Resources	Less CPU power and memory	More CPU power and memory required
Cost		More than Distance vector
Scalability		More scalable than distance vector
Examples	RIP, IGRP	OSPF, IS-IS



Thank You
For Your Attention



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